

# **Time Series Analysis**

## **Lecture Notes and Practical Methods**

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*Comprehensive notes covering fundamental concepts, measurement methods, and  
practical applications of Time Series Analysis*

## Learning Objective

By the end of this lecture, students will be able to:

- Define and identify time series data in real-world scenarios
- Decompose a time series into its four fundamental components
- Apply graphical methods (Freehand and Semi-Average) to measure trend
- Understand basic forecasting models and their applications
- Solve practical problems in time series analysis

## Week 9-10

These notes cover approximately 2 weeks of coursework in Statistics/Data Analysis, focusing on Time Series fundamentals and trend measurement techniques.

# 1 Introduction to Time Series Analysis

## Theoretical Foundations

**Definition:** A time series is a sequence of data points collected for a **single entity** over **regular time intervals**. It represents the evolution of a variable over time.

### 1.1 What is a Time Series?

Time series data is everywhere in our daily lives and professional domains:

- **Personal:** Daily sleep hours tracked by smartwatch, monthly electricity bills
- **Business:** Quarterly sales figures, daily stock prices, monthly website traffic
- **Scientific:** Daily temperature readings, annual rainfall measurements, hourly heart rate data

**Mathematical Representation:** If  $Y$  is the variable of interest and  $t$  represents time, a time series can be represented as:

$$Y_t = \{y_1, y_2, y_3, \dots, y_n\}$$

where  $y_i$  is the observation at time period  $i$ .

### 1.2 Time Series Analysis

Time Series Analysis involves analyzing timestamped data to:

1. Understand underlying patterns and structures
2. Extract meaningful statistics and characteristics
3. Make informed predictions about future values
4. Support decision-making processes

## Applications

### Real-World Applications:

- **Retail:** Predicting future sales to optimize inventory levels
- **Finance:** Forecasting stock prices and commodity prices
- **Agriculture:** Predicting weather patterns for planting/harvesting decisions
- **Healthcare:** Monitoring patient vital signs over time
- **Economics:** Analyzing GDP growth, unemployment rates, inflation

## 2 Components of Time Series

Every time series can be decomposed into four fundamental components. Understanding these is crucial for proper analysis.

### 2.1 1. Trend (T)

The **long-term** direction or pattern in the data over an extended period.

- **Types:** Upward (increasing), Downward (decreasing), or Horizontal (stationary)
- **Characteristics:** Represents persistent, systematic changes
- **Example:** A student's improving grades from 1st to 12th grade
- **Timeframe:** Typically observed over years or decades

**Visual Representation:** A straight or curved line showing overall direction.

### 2.2 2. Seasonality (S)

**Regular, predictable patterns** that repeat over fixed periods (less than one year).

- **Characteristics:** Fixed and known periodicity
- **Examples:**
  - Retail sales peaking every December (holiday season)
  - Ice cream sales increasing every summer
  - Electricity consumption rising every evening
- **Period:** Daily, weekly, monthly, quarterly, or yearly patterns

**Visual Representation:** Repeated peaks and troughs at regular intervals.

### 2.3 3. Cyclical Component (C)

**Long-term fluctuations** without fixed periodicity, often related to economic conditions.

- **Characteristics:** Period varies, typically 2-10 years
- **Examples:** Economic booms and recessions, business cycles
- **Difference from Seasonality:** Not fixed in period, longer duration

**Visual Representation:** Wave-like patterns with varying amplitude and period.

## 2.4 4. Irregular Component (I)

**Random, unpredictable variations** (noise) that cannot be attributed to other components.

- **Characteristics:** Unsystematic, unpredictable, caused by unusual events
- **Examples:** Natural disasters, strikes, sudden policy changes, measurement errors
- **Also called:** Residual, Error, or Noise component

## 2.5 Classical Decomposition Model

The relationship between these components can be expressed as:

$$Y_t = T_t + S_t + C_t + I_t$$

where:

- $Y_t$  = Actual time series value at time  $t$
- $T_t$  = Trend component at time  $t$
- $S_t$  = Seasonal component at time  $t$
- $C_t$  = Cyclical component at time  $t$
- $I_t$  = Irregular component at time  $t$

## 3 Forecasting Models

### 💡 Theoretical Foundations

**Note:** These models are introduced here conceptually. Detailed mathematical treatment is covered in advanced courses.

### 3.1 ARIMA Model

**ARIMA** (AutoRegressive Integrated Moving Average) is one of the most popular time series forecasting models.

- **AR (AutoRegressive):** Uses past values to predict future values

$$\text{AR}(p) : y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \epsilon_t$$

where  $p$  is the order of the AR model.

- **I (Integrated):** Uses differencing to make series stationary (remove trend/seasonality)

$$y'_t = y_t - y_{t-1}$$

- **MA (Moving Average):** Models error as linear combination of past error terms

$$\text{MA}(q) : y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}$$

where  $q$  is the order of the MA model.

### 3.2 Exponential Smoothing

Best for data **without clear trend or seasonality**. Gives more weight to recent observations.

**Simple Exponential Smoothing:**

$$\hat{y}_{t+1} = \alpha y_t + (1 - \alpha)\hat{y}_t$$

where  $\alpha$  is the smoothing parameter ( $0 \leq \alpha \leq 1$ ).

### 3.3 Implementation Tools

- **Python:** Pandas (data manipulation), Matplotlib (visualization), Statsmodels (statistical models)
- **R:** forecast package, tseries package
- **MATLAB:** Econometrics Toolbox
- **Excel:** Built-in forecasting functions

## 4 Measuring Trend: Graphical Methods

### 4.1 Freehand Curve Method

#### Theoretical Foundations

**Principle:** A subjective method where the analyst visually inspects the plotted data and draws a smooth curve that best represents the general direction of the points.

#### 4.1.1 Step-by-Step Procedure

1. **Plot the Data:** On graph paper, mark time on X-axis and variable on Y-axis
2. **Observe Pattern:** Look for overall direction ignoring minor fluctuations
3. **Draw Trend Line:** Draw a smooth line through the center of data points
4. **Ensure Balance:** Total distance of points above line = Total distance below line
5. **Extend for Forecasting:** Extend line to estimate future values

#### 4.1.2 Example Problem

**Given:** Production data (in '000 units) from 1971-1978

| Year       | 1971 | 1972 | 1973 | 1974 | 1975 | 1976 | 1977 | 1978 |
|------------|------|------|------|------|------|------|------|------|
| Production | 50   | 60   | 55   | 58   | 62   | 60   | 57   | 65   |

**Solution Steps:**

1. Plot points: (1971,50), (1972,60), (1973,55), (1974,58), (1975,62), (1976,60), (1977,57), (1978,65)
2. Observe: Overall upward movement despite minor fluctuations

3. Draw a straight line passing through center of data cloud
4. Line should show gentle upward slope

**Advantages:** Simple, quick, flexible, no calculations needed

**Disadvantages:** Subjective, depends on analyst's skill, not precise

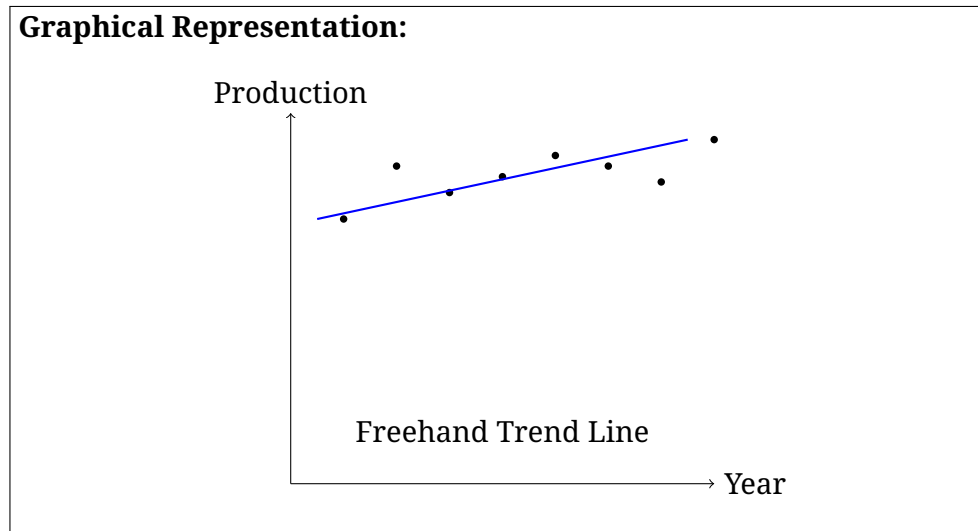


Figure 1: Freehand Method Illustration

## 4.2 Semi-Average Method

### 🔍 Theoretical Foundations

**Principle:** An objective method that divides data into two equal parts, calculates their averages, and uses these to plot a straight trend line.

#### 4.2.1 Case 1: Even Number of Observations

**Example:** Sales data ( Crores) 2012-2019

| Year  | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|-------|------|------|------|------|------|------|------|------|
| Sales | 20   | 22   | 27   | 26   | 28   | 32   | 30   | 35   |

#### Solution Steps:

##### 1. Divide Data: Split into two equal halves

- First half: 2012-2015 (4 years)
- Second half: 2016-2019 (4 years)

##### 2. Calculate Semi-Averages:

$$\text{First Half Average} = \frac{20 + 22 + 27 + 26}{4} = \frac{95}{4} = 23.75$$

$$\text{Second Half Average} = \frac{28 + 32 + 30 + 35}{4} = \frac{125}{4} = 31.25$$

##### 3. Assign to Mid-Points:

- 23.75 assigned to mid-point of first half: between 2013 & 2014 = 2013.5
- 31.25 assigned to mid-point of second half: between 2017 & 2018 = 2017.5

4. **Plot Trend Line:** Connect points (2013.5, 23.75) and (2017.5, 31.25)

#### 4.2.2 Case 2: Odd Number of Observations

**Example:** Data for 7 years (2013-2019, omitting 2012 from previous example)  
**Solution Steps:**

1. **Divide Data:** Omit middle year (2016), split remainder

- First half: 2013, 2014 (2 years)
- Middle (omitted): 2015
- Second half: 2017, 2018 (2 years)

2. **Calculate Semi-Averages:**

$$\text{First Half Average} = \frac{22 + 27}{2} = \frac{49}{2} = 24.5$$

$$\text{Second Half Average} = \frac{32 + 30}{2} = \frac{62}{2} = 31.0$$

3. **Assign to Mid-Points:**

- 24.5 assigned to mid-point of first half: 2013.5
- 31.0 assigned to mid-point of second half: 2017.5

4. **Plot Trend Line:** Connect points (2013.5, 24.5) and (2017.5, 31.0)

#### 4.2.3 Tabular Representation

| Year | Sales | Semi-Total | Semi-Average |
|------|-------|------------|--------------|
| 2012 | 20    | 4*95       | 4*23.75      |
| 2013 | 22    |            |              |
| 2014 | 27    |            |              |
| 2015 | 26    |            |              |
| 2016 | 28    | 4*125      | 4*31.25      |
| 2017 | 32    |            |              |
| 2018 | 30    |            |              |
| 2019 | 35    |            |              |

**Advantages:** Objective, simple calculations, reproducible

**Disadvantages:** Assumes linear trend, sensitive to extreme values

## 5 Comparative Analysis of Methods

| Method                | Key Principle                                      | Advantages                                               | Limitations                                 |
|-----------------------|----------------------------------------------------|----------------------------------------------------------|---------------------------------------------|
| <b>Freehand</b>       | Visual inspection and manual curve fitting         | Simple, quick, flexible, no calculations                 | Subjective, depends on analyst's skill      |
| <b>Semi-Average</b>   | Dividing data into halves and using averages       | Objective, simple calculations, reproducible             | Assumes linearity, affected by extremes     |
| <b>Moving Average</b> | Smoothing using averages of subsets (future topic) | Reduces randomness, identifies trend clearly             | Loss of data points, lag in identification  |
| <b>Least Squares</b>  | Mathematical minimization of errors (future topic) | Most accurate, provides equation, statistical properties | Complex calculations, assumes specific form |

## 6 Practice Problems

### 6.1 Problem 1: Component Identification

A company's monthly data shows:

- Steady growth over 5 years
- Peaks every December
- A 2-year period of slow growth during economic recession
- Sudden drop in one month due to workers' strike

Identify the four time series components in this description.

### 6.2 Problem 2: Freehand Method Application

Given the following profit data (in Lakhs):

|        |      |      |      |      |      |      |      |
|--------|------|------|------|------|------|------|------|
| Year   | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 |
| Profit | 30   | 35   | 32   | 40   | 38   | 45   | 42   |

Plot the data and draw a freehand trend line. Comment on the trend direction.

### 6.3 Problem 3: Semi-Average Method (Even Data)

Calculate trend using Semi-Average method:

|            |      |      |      |      |      |      |      |      |
|------------|------|------|------|------|------|------|------|------|
| Year       | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 |
| Production | 100  | 110  | 115  | 105  | 125  | 130  | 120  | 135  |

### 6.4 Problem 4: Semi-Average Method (Odd Data)

Apply Semi-Average method to:

|         |      |      |      |      |      |      |      |
|---------|------|------|------|------|------|------|------|
| Year    | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
| Revenue | 80   | 85   | 90   | 82   | 95   | 98   | 93   |

### 6.5 Problem 5: Trend Interpretation

The semi-averages for a 10-year profit series are:

- Point A: (Year 2015.5, Average = 52 Crores)
- Point B: (Year 2020.5, Average = 68 Crores)



- (a) What is the annual increase suggested by this trend?
- (b) Estimate the profit for 2023.
- (c) If actual 2021 profit was 70 Crores, calculate the irregular component.

## 7 Solutions to Practice Problems

### Solution 1: Component Identification

- **Trend:** Steady growth over 5 years
- **Seasonality:** Peaks every December
- **Cyclical:** 2-year period of slow growth during recession
- **Irregular:** Sudden drop due to strike

### Solution 2: Freehand Method

1. Plot points: (2015,30), (2016,35), (2017,32), (2018,40), (2019,38), (2020,45), (2021,42)
2. Draw a line through center showing upward trend
3. Comment: Overall increasing trend with some fluctuations

### Solution 3: Semi-Average (Even)

$$\begin{aligned}\text{First Half Average} &= \frac{100 + 110 + 115 + 105}{4} = \frac{430}{4} = 107.5 \\ \text{Second Half Average} &= \frac{125 + 130 + 120 + 135}{4} = \frac{510}{4} = 127.5 \\ \text{Trend Line: Connect } &(2015.5, 107.5) \text{ and } (2019.5, 127.5)\end{aligned}$$

### Solution 4: Semi-Average (Odd)

$$\begin{aligned}\text{First Half Average} &= \frac{80 + 85}{2} = 82.5 \quad (\text{Mid-point: } 2016.5) \\ \text{Second Half Average} &= \frac{95 + 98}{2} = 96.5 \quad (\text{Mid-point: } 2020.5) \\ \text{Trend Line: Connect } &(2016.5, 82.5) \text{ and } (2020.5, 96.5)\end{aligned}$$

### Solution 5: Trend Interpretation

- (a) Annual increase =  $\frac{68-52}{2020.5-2015.5} = \frac{16}{5} = 3.2$  Crores per year
- (b) For 2023:  $68 + 2.5 \times 3.2 = 68 + 8 = 76$  Crores
- (c) Irregular component = Actual - Trend =  $70 - (68 + 0.5 \times 3.2) = 70 - 69.6 = 0.4$  Crores

## 📖 Book Chapter

### Next Chapter Preview:

- Moving Average Method (3-year, 4-year, 5-year) • Least Squares Method (linear trend fitting) • Seasonal indices calculation • Advanced forecasting techniques

## 📖 Learning Resources

### Recommended Resources:

- **Textbook:** "Business Statistics" by Gupta and Gupta
- **Software:** Python with Pandas/Matplotlib, Excel
- **Online:** Khan Academy Statistics, Coursera Time Series courses
- **Practice:** Real-world datasets from Kaggle, government statistics portals

## 8 Mathematical Foundations of Time Series

### 💡 Theoretical Foundations

A **Time Series** is a realization of a stochastic process

$$\{X_t\}_{t \in T}$$

where  $T = \{0, 1, 2, \dots\}$  represents time.

Unlike independent data, time series observations are dependent:

$$\text{Cov}(X_t, X_{t-k}) \neq 0$$

### 8.1 Weak Stationarity

A process is weakly stationary if:

1. Constant Mean

$$E[X_t] = \mu$$

2. Constant Variance

$$\text{Var}(X_t) = \sigma^2 < \infty$$

3. Covariance depends only on lag:

$$\text{Cov}(X_t, X_{t+h}) = \gamma(h)$$

## 8.2 White Noise Process

### 💡 Theoretical Foundations

White Noise is defined as

$$\epsilon_t \sim WN(0, \sigma^2)$$

such that:

$$E(\epsilon_t) = 0$$

$$Var(\epsilon_t) = \sigma^2$$

$$Cov(\epsilon_t, \epsilon_s) = 0, \quad t \neq s$$

## 9 Linear Time Series Models

### 9.1 Autoregressive Model AR(p)

$$X_t = c + \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + \epsilon_t$$

**Stationarity Condition**

Characteristic equation:

$$1 - \phi_1 z - \phi_2 z^2 - \dots - \phi_p z^p = 0$$

The AR model is stationary if all roots satisfy:

$$|z| > 1$$

Given:

$$X_t = 0.6X_{t-1} + \epsilon_t$$

If  $X_{t-1} = 10$  and  $\epsilon_t = 2$ :

$$X_t = 0.6(10) + 2 = 8$$

### 9.2 Variance of AR(1)

Formula:

$$Var(X_t) = \frac{\sigma^2}{1 - \phi^2}$$

Given  $\phi = 0.5$  and  $\sigma^2 = 9$

$$Var(X_t) = \frac{9}{1 - 0.25} = 12$$

### 9.3 Moving Average Model MA(q)

$$X_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \dots + \theta_q \epsilon_{t-q}$$

All MA(q) models are stationary.

Given:

$$X_t = \epsilon_t + 0.4\epsilon_{t-1}$$

If  $\epsilon_t = 5$ ,  $\epsilon_{t-1} = 3$

$$X_t = 5 + 1.2 = 6.2$$

## 9.4 Mean and Variance of MA(1)

$$E(X_t) = 0$$

$$Var(X_t) = \sigma^2(1 + \theta^2)$$

If  $\theta = 0.5$  and  $\sigma^2 = 4$

$$Var(X_t) = 4(1.25) = 5$$

## 10 Wold Decomposition Theorem

### 💡 Theoretical Foundations

Every stationary process can be represented as:

$$X_t = \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j} + \eta_t$$

This means every stationary process is an infinite moving average.

## 11 Model Identification

### 11.1 Autocorrelation Function (ACF)

Measures correlation between:

$$Corr(X_t, X_{t-k})$$

### 11.2 Partial Autocorrelation (PACF)

Measures correlation after removing intermediate lags.

| Model | ACF             | PACF            |
|-------|-----------------|-----------------|
| AR(p) | Tails off       | Cuts off at $p$ |
| MA(q) | Cuts off at $q$ | Tails off       |

## 12 Model Selection Using AIC

$$AIC = -2 \ln(\hat{L}) + 2k$$

Model Comparison:

| Model | Log Likelihood | Parameters |
|-------|----------------|------------|
| AR(1) | -120           | 2          |
| AR(2) | -110           | 3          |

$$AIC_{AR(1)} = 244$$

$$AIC_{AR(2)} = 226$$

Thus AR(2) is better.

## 13 Classical Components of Time Series

Time series consists of:

$$Y_t = T_t + S_t + C_t + I_t$$

- Trend
- Seasonal
- Cyclical
- Irregular

## 14 Forecasting Models

### 14.1 ARIMA Model

Combines AR, differencing, and MA models.

### 14.2 Exponential Smoothing

$$\hat{y}_{t+1} = \alpha y_t + (1 - \alpha)\hat{y}_t$$

## 15 Trend Measurement Methods

### 15.1 Freehand Method

Plot data and draw smooth curve visually.

### 15.2 Semi-Average Method

Divide data into two equal parts and compute averages.

Sales Data:

20 22 27 26 28 32 30 35

First average:

23.75

Second average:

31.25

## 16 Practice Problems

1. Compute  $X_t$  if:

$$X_t = 0.3X_{t-1} + \epsilon_t$$

with  $X_{t-1} = 20$  and  $\epsilon_t = 1$ .

2. Check stationarity if  $\phi = 0.95$ .
3. For MA(1) with  $\theta = 0.7$ ,  $\sigma^2 = 2$ , find variance.

## 17 Solutions

### Problem 1

$$X_t = 0.3(20) + 1 = 7$$

### Problem 2

Since  $|\phi| < 1$ , stationary.

### Problem 3

$$\text{Var}(X_t) = 2(1 + 0.49) = 2.98$$

## 18 Summary

- Time series models describe dependence over time.
- Stationarity is essential for modeling.
- AR and MA are fundamental models.
- ACF and PACF help model identification.
- AIC helps select best model.
- Time series analysis studies data collected over regular time intervals • Four components: Trend (long-term direction), Seasonality (regular patterns), Cyclical (economic waves), Irregular (random noise) • Freehand method is subjective but quick; Semi-average method is objective and calculative • Choice of method depends on data characteristics and analysis requirements • Proper trend identification is crucial for accurate forecasting and decision-making